

Data Quality Issues - Master Reference

For Data Quality Framework / Product Design

This document catalogs data quality issues encountered across data engineering pipelines, organized by dimension (nature). It is intended as a working reference to design rule libraries, checks, and scoring dimensions for a DQ framework/product.

1. Completeness Issues

Data that should exist is missing.

#	Issue	Description / Example
1.1	Missing/Null values	Mandatory field (e.g., customer_id) is NULL
1.2	Blank/empty string vs. true NULL	"" stored instead of NULL, causing inconsistent handling
1.3	Missing records	Expected rows absent (e.g., a store with zero sales rows on a business day)
1.4	Missing files/partitions	An expected daily file/partition never arrived
1.5	Partial file load	File truncated mid-load; only part of the data ingested
1.6	Missing mandatory attributes	Required column exists in schema but not populated for all records
1.7	Missing relationships	Parent record exists but child/detail records missing (e.g., order with no order lines)
1.8	Missing historical data	Gaps in time-series data (e.g., missing month in a 5-year trend)
1.9	Default value masking missing data	Nulls silently replaced with 0, "N/A", "9999-12-31" hiding true absence
1.10	Missing reference/lookup data	Code exists in transaction table but missing in dimension/lookup table

2. Accuracy / Correctness Issues

Data is present but does not reflect the true/real-world value.

#	Issue	Description / Example
2.1	Incorrect values	Wrong data entered (e.g., age = 250)
2.2	Outdated/stale values	Address not updated after customer moved
2.3	Miscalculated derived fields	Total $\neq$ sum of line items due to formula error

2.4	Unit mismatch	Weight recorded in kg vs lbs without conversion
2.5	Precision/rounding errors	Currency truncated instead of rounded, causing penny discrepancies
2.6	Wrong data source mapped	Field populated from incorrect upstream source/column
2.7	Transcription/typo errors	Manual entry errors (“Newyrok” instead of “New York”)
2.8	Sensor/device measurement errors	Faulty IoT sensor sending incorrect readings
2.9	Currency/exchange rate errors	Using stale FX rate for conversion
2.10	Incorrect classification/categorization	Product mapped to wrong category/hierarchy
2.11	Business logic errors in transformation	ETL logic incorrectly implements business rule

3. Consistency Issues

Same data conflicts across fields, records, tables, or systems.

#	Issue	Description / Example
3.1	Cross-field inconsistency	ship_date earlier than order_date
3.2	Cross-table inconsistency	Customer status = “Active” in CRM but “Closed” in Billing
3.3	Cross-system inconsistency	Same KPI computed differently in two BI tools
3.4	Inconsistent aggregation logic	Monthly total $\neq$ sum of daily totals due to differing rules
3.5	Redundant data mismatch	Same attribute (e.g., email) stored in 2 tables with different values
3.6	Temporal inconsistency	Same entity has conflicting states at same timestamp (SCD conflict)
3.7	Currency/unit inconsistency across records	Same metric reported in different units within same dataset
3.8	Inconsistent grain	Table mixes daily and monthly grain rows without a discriminator
3.9	Inconsistent code/value mapping	“Y/N” in one system vs “1/0” in another for the same concept

4. Validity / Conformity Issues

Data violates defined format, domain, or business rule.

#	Issue	Description / Example
4.1	Format violation	Date stored as “13/45/2023”
	Data type	Numeric field contains alphabetic

4.2	mismatch	characters
4.3	Domain/range violation	Discount % = 150%
4.4	Pattern/regex violation	Email without "@", invalid phone number format
4.5	Invalid enumeration value	Status = "Pnding" instead of allowed set {Pending, Approved, Rejected}
4.6	Business rule violation	Order shipped before payment confirmed
4.7	Referential/lookup code invalid	Country code "ZZ" not in ISO list
4.8	Length constraint violation	Field exceeds/undercuts expected character length
4.9	Encoding violation	Non-UTF8 characters causing garbled text (mojibake)
4.10	Case sensitivity violation	Enum expected uppercase but stored mixed-case
4.11	Cross-field validity rule violation	"Age" = 5 but "Occupation" = "Retired"

5. Uniqueness / Duplication Issues

Same real-world entity represented more than once, or key uniqueness violated.

#	Issue	Description / Example
5.1	Exact duplicate records	Same row inserted twice
5.2	Fuzzy/near duplicates	"Jon Smith" vs "John Smith" - same person, different spelling
5.3	Primary key violation	Duplicate primary key values in table meant to be unique
5.4	Duplicate due to reprocessing	Pipeline re-run without idempotency causing double counts
5.5	Duplicate across merge/union sources	Same record ingested from two overlapping source systems
5.6	Composite key duplication	Uniqueness violated only when checked across multiple columns together
5.7	Duplicate due to late-arriving data reprocessing	Backfill logic re-inserts already-processed records

6. Timeliness / Freshness Issues

Data is not available or updated within the required time window.

#	Issue	Description / Example
---	-------	-----------------------

6.1	Stale data	Dashboard shows yesterday's data when today's was expected
6.2	Late-arriving data	Source system sends data past agreed SLA cutoff
6.3	Out-of-order events	Event with earlier timestamp arrives after a later one (stream processing)
6.4	Delayed batch jobs	ETL job delayed, downstream dependent jobs run on old data
6.5	Missed SLA / refresh window	Data not refreshed per contractually defined frequency
6.6	Clock/timezone drift	Source and target systems have unsynced clocks causing timestamp errors
6.7	Real-time vs batch latency mismatch	Streaming layer ahead of batch layer, causing reconciliation gaps

7. Integrity / Referential Issues

Relationships between data entities are broken.

#	Issue	Description / Example
7.1	Orphan records	Foreign key references non-existent parent (e.g., order with invalid customer_id)
7.2	Broken referential integrity after deletes	Parent deleted, child records remain (dangling references)
7.3	Circular references	A references B, B references A causing infinite loop in hierarchy
7.4	Hierarchy/parent-child violations	Employee reports to self, or org hierarchy has a loop
7.5	Cardinality violation	One-to-one relationship has multiple matches
7.6	Cross-database referential mismatch	Foreign key in one DB doesn't resolve in another linked DB
7.7	Slowly Changing Dimension (SCD) integrity issues	Multiple "current = Y" flags for same key

8. Structural / Schema Issues

Issues in the shape, definition, or evolution of data structures.

#	Issue	Description / Example
8.1	Schema drift	Source adds/removes/renames a column without notice
8.2	Data type drift	Column type changes from INT to STRING upstream
8.3	Unexpected/extra columns	New unmapped column appears in feed
8.4	Column order dependency failure	Pipeline assumes fixed column order; source reorders columns

8.5	Nested/complex structure issues	Malformed JSON/XML, broken nesting, inconsistent array lengths
8.6	Incompatible schema evolution	Backward-incompatible Avro/Parquet schema change breaks readers
8.7	Flattening errors	Incorrect flattening of nested structures during transformation
8.8	Metadata schema mismatch	Data catalog schema doesn't match actual table structure
8.9	Header/footer row issues in files	File missing header, or header row treated as data
8.10	Delimiter/parsing issues	Comma within a CSV field not escaped, causing column shift

9. Format / Standardization Issues

Data lacks a consistent representation standard.

#	Issue	Description / Example
9.1	Inconsistent date/time formats	"MM/DD/YYYY" vs "DD-MM-YYYY" mixed in same field
9.2	Inconsistent naming conventions	"NY", "N.Y.", "New York" for the same value
9.3	Inconsistent casing	"MALE", "male", "Male" not standardized
9.4	Inconsistent address formats	Free text address vs structured address fields
9.5	Whitespace/trailing space issues	" John " vs "John" treated as different values
9.6	Special character handling	Unescaped quotes, tabs, or newlines within fields
9.7	Locale-specific formatting issues	Decimal comma (1.234,56) vs decimal point (1,234.56) confusion
9.8	Non-standardized units of measure	"kg", "Kg", "KG", "kilograms" used interchangeably

10. Volume / Reconciliation Issues

Record counts or totals don't match expectations across stages/systems.

#	Issue	Description / Example
10.1	Row count mismatch (source vs target)	Source has 10,000 rows, target loaded only 9,850
10.2	Sudden volume spike/drop	Daily load volume deviates significantly from historical pattern
10.3	Control total mismatch	Sum of a financial column in source $\neq$ sum in target
10.4	Fan-out/fan-in errors in joins Data loss	Unintended row multiplication due to join cardinality issues

10.5	during transformation	Filter/dedup logic drops valid records unintentionally
10.6	Data loss during transmission	Network/file transfer truncation
10.7	Reconciliation break between layers	Bronze vs Silver vs Gold layer counts don't tie out (lakehouse)

11. Statistical / Distributional / Anomaly Issues

Data deviates from expected statistical behavior (often caught via profiling/ML-based checks).

#	Issue	Description / Example
11.1	Outliers	Transaction amount = \$10,000,000 for a retail purchase
11.2	Unexpected distribution shift (data drift)	Category proportions shift drastically vs historical baseline
11.3	Skewed/imbalanced data	One category dominates 99% of records unexpectedly
11.4	Concept drift (for ML-consuming pipelines)	Relationship between features and target changes over time
11.5	Anomalous null rate spike	Null percentage in a field jumps from 1% to 40% suddenly
11.6	Cardinality anomaly	Unique value count in a categorical column changes abruptly
11.7	Seasonal pattern break	Expected seasonal trend (e.g., holiday spike) doesn't occur
11.8	Correlation break	Two historically correlated metrics suddenly diverge

12. Metadata / Lineage / Documentation Issues

Issues in the data describing the data.

#	Issue	Description / Example
12.1	Missing/incorrect data dictionary	Column meaning undocumented or mislabeled
12.2	Broken/missing lineage	Unable to trace field back to its source system
12.3	Undocumented transformation logic	Business logic embedded in code with no documentation
12.4	Stale metadata	Catalog says "daily refresh" but job now runs weekly
12.5	Ownership/stewardship gaps	No identified data owner for critical dataset
12.6	Ambiguous business definitions	"Active customer" defined differently by different teams No version control on

12.7	Versioning issues	schema/reference data changes
12.8	Undocumented assumptions in derived fields	Calculated KPI logic not traceable to a rule

13. Security, Privacy & Compliance-Related DQ Issues

Data quality problems with regulatory/governance implications.

#	Issue	Description / Example
13.1	PII/PHI exposed in non-secure fields	SSN stored in plain text in an unrestricted table
13.2	Improper masking/anonymization	Masking logic reversible or incomplete
13.3	Access control mismatch	Sensitive data accessible to unauthorized roles
13.4	Consent/data usage violations	Customer data used beyond consented purpose
13.5	Retention policy violations	Data retained beyond legal/contractual limits
13.6	Cross-border data residency issues	Data replicated to a region violating data sovereignty rules
13.7	Audit trail gaps	Changes to sensitive data not logged

14. Data Engineering / Pipeline-Specific Technical Issues

Issues rooted in ETL/ELT engineering, infrastructure, and processing logic.

#	Issue	Description / Example
14.1	Job failure / partial execution	Pipeline fails midway, leaving target in inconsistent state
14.2	Non-idempotent pipeline reruns	Re-running a job duplicates or corrupts data
14.3	Incorrect incremental/CDC logic	Change Data Capture misses updates/deletes
14.4	Watermark/checkpoint errors (streaming)	Incorrect watermarking causes dropped or duplicate events
14.5	Join key mismatch during transformation	Join on mismatched data types/formats silently drops rows
14.6	Incorrect null handling in transformations	COALESCE/default logic masks real nulls
14.7	Time zone conversion errors in pipelines	UTC vs local time mishandled during transformation
14.8	Encoding/decoding errors	UTF-8 vs Latin-1 mismatch corrupts text
14.9	Race conditions in parallel processing	Concurrent writes cause inconsistent state
14.10	Configuration drift between	Dev/QA/Prod pipeline configs differ, causing inconsistent output

	environments	
14.11	Silent failures (no error thrown but wrong output)	Transformation completes “successfully” with wrong logic
14.12	API/data contract breaking changes	Upstream API changes response structure without version bump
14.13	Resource/timeout-induced data truncation	Job times out and writes partial output without failing loudly
14.14	Improper error handling / swallowed exceptions	Errors caught but not logged/alerted, issue goes unnoticed
14.15	Backfill errors	Historical reprocessing applies current logic incorrectly to past data

15. Cross-System / Interoperability Issues

Issues arising when multiple systems, vendors, or formats interact.

#	Issue	Description / Example
15.1	Master data mismatch across systems	Same product has different IDs in ERP vs e-commerce platform
15.2	API version mismatch	Consumer expects v1 payload, producer sends v2
15.3	File format incompatibility	Producer sends Parquet, consumer expects CSV
15.4	Character set/encoding mismatch between systems	Source uses EBCDIC, target expects ASCII/UTF-8
15.5	Third-party/vendor data quality issues	External data provider sends inconsistent or delayed feeds
15.6	Inconsistent business calendars	Fiscal year definitions differ between finance and operations systems

16. Governance & Process Issues

Organizational/process root causes that manifest as data quality problems.

#	Issue	Description / Example
16.1	No single source of truth	Multiple systems considered authoritative for the same entity
16.2	Lack of data quality SLAs	No agreed thresholds for acceptable error rates
16.3	No change management for reference data	Codes/mappings updated without notifying downstream consumers

16.4	Manual data entry dependency	High reliance on manual processes prone to human error
16.5	Lack of validation at point of entry	Source application allows free-text entry with no validation
16.6	Undefined data quality ownership/RACI	No clear accountability when issues are found
16.7	Reactive-only issue detection	Issues found by business users instead of proactive monitoring

17. Change Management / Release & Deployment Issues

Data quality failures caused by how code, pipelines, or frameworks are changed and shipped — not by the data content itself. These are a leading root cause of DQ incidents in mature platforms even though they originate in engineering process, not data.

#	Issue	Description / Example
17.1	No regression testing before deployment	A pipeline/framework upgrade ships without running existing validation checks, silently breaking downstream logic
17.2	Breaking change deployed without a validation gate	New transformation logic goes live without passing through a staging/UAT environment
17.3	CI/CD deployment failure affecting data availability	A failed merge/deploy leaves a pipeline job in a broken or partially-updated state
17.4	Framework/platform upgrade side effects	Upgrading a shared ingestion framework (e.g., an internal ETL tool) changes behavior for all dependent pipelines without notice
17.5	Untested hotfixes applied directly to production	Emergency fix bypasses normal review/testing, introducing a new defect
17.6	Environment parity gaps	Logic behaves differently in dev/staging vs. production due to config or data differences, so tests pass but production fails
17.7	No rollback plan / inability to revert	A bad deployment corrupts data and there is no clean way to restore the prior state
17.8	Dependency/library version drift	An updated library or package changes parsing/formatting behavior unexpectedly (e.g., date parsing library update)
17.9	Uncoordinated multi-team releases	Two teams deploy dependent changes (e.g., schema change + consuming pipeline change) out of sync with each other

18. Infrastructure / Performance & Capacity Issues (Adjacent — Not Core DQ)

Not data-content quality issues in the classical sense, but operational failures that produce the same symptom users report as “bad data” (missing, late, or unavailable data). Worth tracking alongside DQ because they share detection/alerting infrastructure, even though root cause and remediation owner differ.

#	Issue	Description / Example
18.1	Compute capacity exhaustion	Cluster/warehouse under-provisioned, causing job timeout or failure during peak load
18.2	Query/dashboard performance degradation	Report times out or fails to render due to unoptimized queries or growing data volume
18.3	Storage/quota limits reached	Job fails because target storage or table quota is exceeded
18.4	Concurrency/contention bottlenecks	Multiple jobs/users competing for the same resource cause delays or failures
18.5	Network/connectivity failures	Transient network issues interrupt ingestion or file transfer, mimicking a “missing data” issue
18.6	Cost-driven throttling	Auto-scaling or budget limits throttle a job, causing it to run slower or fail to complete within SLA

Mapping to Classic DQ Dimensions (for framework scoring)

Dimension	Sections Covered
Completeness	Section 1
Accuracy	Section 2
Consistency	Section 3
Validity	Section 4
Uniqueness	Section 5
Timeliness	Section 6
Integrity	Section 7
Structural Conformance	Section 8
Standardization	Section 9
Reconciliation/Volume	Section 10
Statistical Reliability	Section 11
Metadata Quality	Section 12
Security/Compliance	Section 13
Engineering Reliability	Section 14
Interoperability	Section 15
Governance	Section 16
Change Management / Release Risk	Section 17
Infrastructure / Performance (adjacent)	Section 18

Suggested Use in Framework Design

- Use Sections 1-11 as the **rule engine dimensions** (things you write automated checks for: null checks, regex, range, uniqueness,

referential, freshness, drift, reconciliation, anomaly detection).

- Use Section 12 (Metadata) to drive **catalog/lineage completeness scoring**.
- Use Section 13 (Security/Compliance) as a **separate risk-scoring dimension**, since severity/impact differs from standard DQ dimensions.
- Use Section 14 (Pipeline-specific) to drive **operational health monitoring** (job success, SLA, idempotency checks) — distinct from data-content quality.
- Use Section 15 (Interoperability) for **contract-testing** between systems (schema/API contract validation).
- Use Section 16 (Governance) to drive **process/maturity assessment**, not automated rules — useful for a DQ maturity scorecard alongside the technical checks.
- Use Section 17 (Change Management/Release) to drive **pre-deployment gates**: regression suites, staging validation, and rollback readiness checks tied to your CI/CD pipeline rather than the data pipeline itself.
- Track Section 18 (Infrastructure/Performance) on the **same monitoring dashboard** as DQ metrics — since users experience them identically (“my data is missing/late”) — but keep them in a separate root-cause category so remediation routes to Platform/Infra rather than Data Engineering.